VTPEH 6270 – Final Report: Income, Physical Inactivity, and Obesity Prevalence in U.S. Adults

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Contents

1 Introduction

1.1 Research Questions

This final report addresses one central research question using 2011–2024 CDC Behavioral Risk Factor Surveillance System (BRFSS) data:

If a state has low obesity, is that because its residents have higher incomes, are more physically active, or both?

This is examined in three parts:

1. Is obesity prevalence associated with income category among U.S. adults, such that higher income categories have lower obesity prevalence?
2. What is the state-level relationship between income and obesity – do higher-income states tend to have lower obesity rates?
3. How does physical inactivity relate to state-level obesity prevalence, and does the association persist after controlling for income?

1.2 Scientific Plausibility and Context

Obesity is a major and growing public health burden in the United States, affecting more than 40% of adults and contributing substantially to chronic disease burden, healthcare costs, and premature mortality (Centers for Disease Control and Prevention, 2022). Understanding modifiable behavioural and socioeconomic determinants of obesity is therefore central to designing effective prevention strategies.

Income and obesity. A robust socioeconomic gradient in obesity has been documented across multiple national surveys. Lower-income adults consistently exhibit higher rates of obesity, a pattern attributed to differential access to nutritious food, recreational facilities, and health-promoting environments (Ogden et al., 2017). The energy-dense, low-cost food environments that characterise lower-income neighbourhoods are thought to drive excess caloric intake, while high work-hour burdens leave limited time for physical activity (Drewnowski & Specter, 2004).

Physical inactivity and obesity. Energy balance theory predicts that insufficient physical activity relative to caloric intake leads to adipose tissue accumulation over time (Hill et al., 2003). At the population level, Ding et al. (2016) documented large population-attributable fractions of physical inactivity for obesity globally, and state-level analyses show that regions with higher sedentary behaviour tend to have higher obesity rates.

Joint pathways. Income and physical inactivity are not independent: lower-income individuals face structural barriers to exercise (lack of safe parks, gym access, leisure time), suggesting that the income–obesity gradient may operate partly *through* physical inactivity (Katzmarzyk et al., 2000). This motivates a joint analysis that simultaneously considers both factors at the state level.

Based on this background, we hypothesise that: (a) lower income categories will be associated with higher obesity prevalence; (b) states with lower average incomes will have higher obesity rates; and (c) physical inactivity will be a significant positive predictor of state-level obesity even after controlling for income.

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2 Material & Methods

2.1 Data

Data were obtained from the CDC's **Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System (BRFSS)** dataset, publicly available via the CDC Open Data portal (https://catalog.data.gov/dataset/nutrition-physical-activity-and-obesity-behavioral-risk-factor-surveillance-system). The BRFSS is an annual telephone survey of non-institutionalised U.S. adults (>=18 years) conducted in all 50 states, the District of Columbia, and U.S. territories (Centers for Disease Control and Prevention, 2023). The full dataset spans **2011–2024** and contains 110,880 state-year-stratification records.

2.1.1 Variables

Table 1: Analytic variables and their roles.

Variable	Description	Role
pct_obese	% adults aged >=18 with obesity (BMI >=30) – state-year-income or Total stratum	Outcome
pct_inactive	% adults reporting no leisure-time physical activity – state-year Total stratum	Predictor
income_step	Ordered income step (1 = <$15k; 6 = >=$75k)	Predictor
YearStart	Survey year (2011-2024)	Covariate / grouping
LocationDesc	U.S. state or territory	Grouping

2.1.2 Data Cleaning and Transformation

```
## Income-stratified rows : 4505

## State-year analytic rows: 750

## States/territories      : 55

## Years covered           : 2011 - 2024
```

The following cleaning steps were applied. First, subsetting: for Q1 only Income-stratum rows were retained; for Q2–Q3 only Total-stratum rows were used to avoid double-counting. Second, income proxy: a sample-size-weighted mean income step (1–6) was computed per state-year and merged in. Third, missing values: rows with Data_Value_Footnote (suppressed estimates) were excluded; remaining NAs removed via complete-case analysis. No transformations were required as both outcomes are on a 0–100 percentage scale with approximately linear relationships to predictors.

2.2 Statistical Analyses

The analysis proceeds in three blocks. Block 1 (Q1) addresses the income gradient in obesity through descriptive summaries and visualisations across six ordered income categories, quantified by simple linear regression of obesity on income_step. Block 2 (Q2) uses the state-level income proxy in a Pearson correlation and simple linear regression. Block 3 (Q3) examines physical inactivity via Pearson correlation, simple linear regression, and a multiple regression adding year and income proxy as covariates. Assumption checks (Shapiro-Wilk, residuals vs. fitted, Q-Q plot) are reported for each model.

2.3 Code Repository

All data, R Markdown source, and scripts are available at: https://github.com/mphashi/FinalProject-VTPEH-6270

Exploration of data available at ShinyApp: https://mphashita.shinyapps.io/RFINALSHINYAPP/

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3 Results

3.1 Descriptive Statistics

Table 2: Descriptive statistics for obesity prevalence (%) by income category (BRFSS 2011-2024).

Income Category	N	Mean (%)	SD	Median	Q25	Q75
Less than $15,000	751	35.8	5.2	35.9	32.5	39.3
$15,000 - $24,999	751	34.7	5.0	34.6	31.5	37.9
$25,000 - $34,999	751	33.3	5.3	33.2	29.6	36.7
$35,000 - $49,999	751	32.9	4.9	33.0	29.4	36.2
$50,000 - $74,999	750	32.3	5.0	32.0	28.7	35.9
$75,000 or greater	751	29.0	6.0	28.5	24.5	33.1

Table 3: Descriptive statistics for state-year physical inactivity and obesity prevalence (Total stratum, 2011-2024).

N (state-year obs)	Mean inactivity (%)	SD inactivity	Mean obesity (%)	SD obesity
750	24.7	4.9	31	4.3

Across the income-stratified data, mean obesity prevalence decreases from **35.8%** in the lowest income group to **29%** in the highest. At the state-year level, mean physical inactivity is 24.7% (SD = 4.9) and mean obesity is 31% (SD = 4.3).

3.2 Visualisation

3.2.1 Q1 – Income Gradient in Obesity

3.2.2 Q2 – State-Level Income Proxy and Obesity

3.2.3 Q3 – Physical Inactivity vs. Obesity

3.3 Assumption Checks

Table 4: Shapiro-Wilk test of residual normality.

Model	W statistic	p-value
Income step -> Obesity (income-stratified)	0.9944	3.76e-12
Physical inactivity -> Obesity (state-level)	0.9928	1.14e-03

Both Shapiro-Wilk tests yield $p < 0.05$; however, with large sample sizes ($n = 4505$ for the income model; $n = 750$ for the state-level model), this reflects minor departures from normality rather than a critical violation. Q-Q plots confirm approximate normality in the central range, and inference is robust by the Central Limit Theorem. Residuals-vs-fitted plots show no strong systematic curvature, supporting linearity and homoscedasticity.

3.4 Statistical Analyses

3.4.1 Q1 – Income Gradient in Obesity: Simple Linear Regression

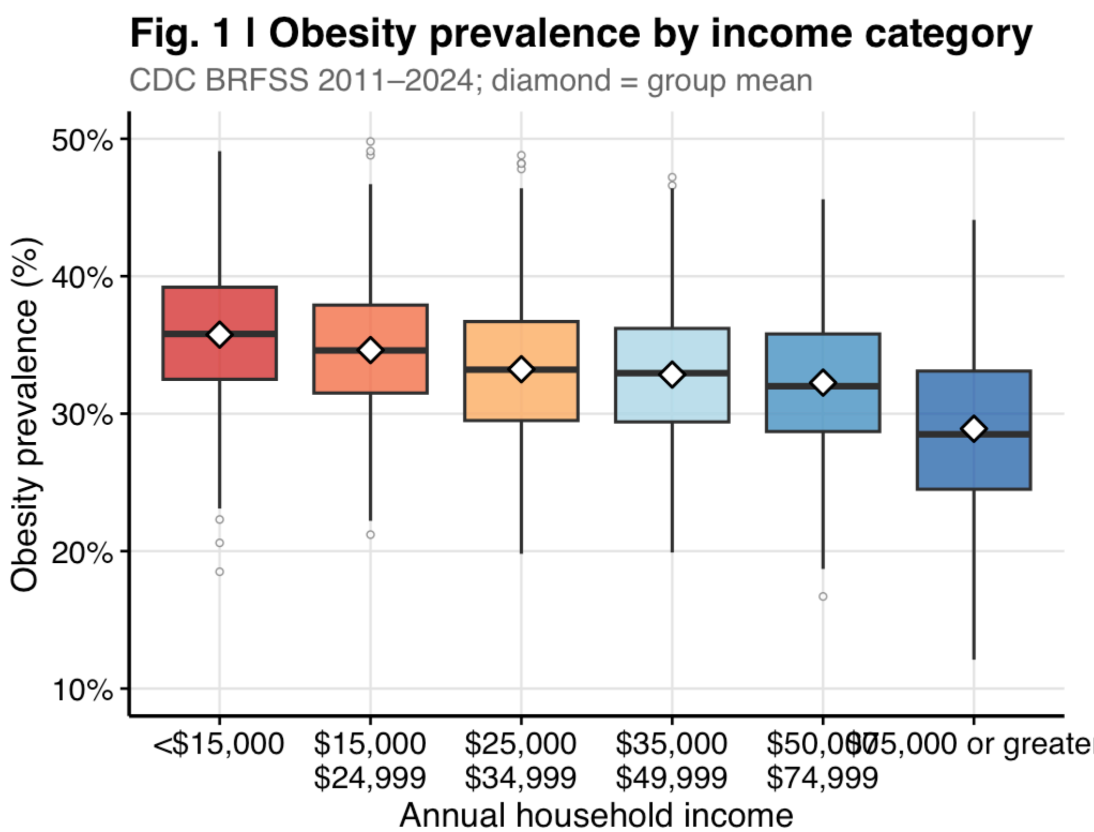

Figure 1: Fig. 1 | Distribution of obesity prevalence by income category. Diamond = group mean; boxes show median ± IQR. CDC BRFSS 2011–2024.

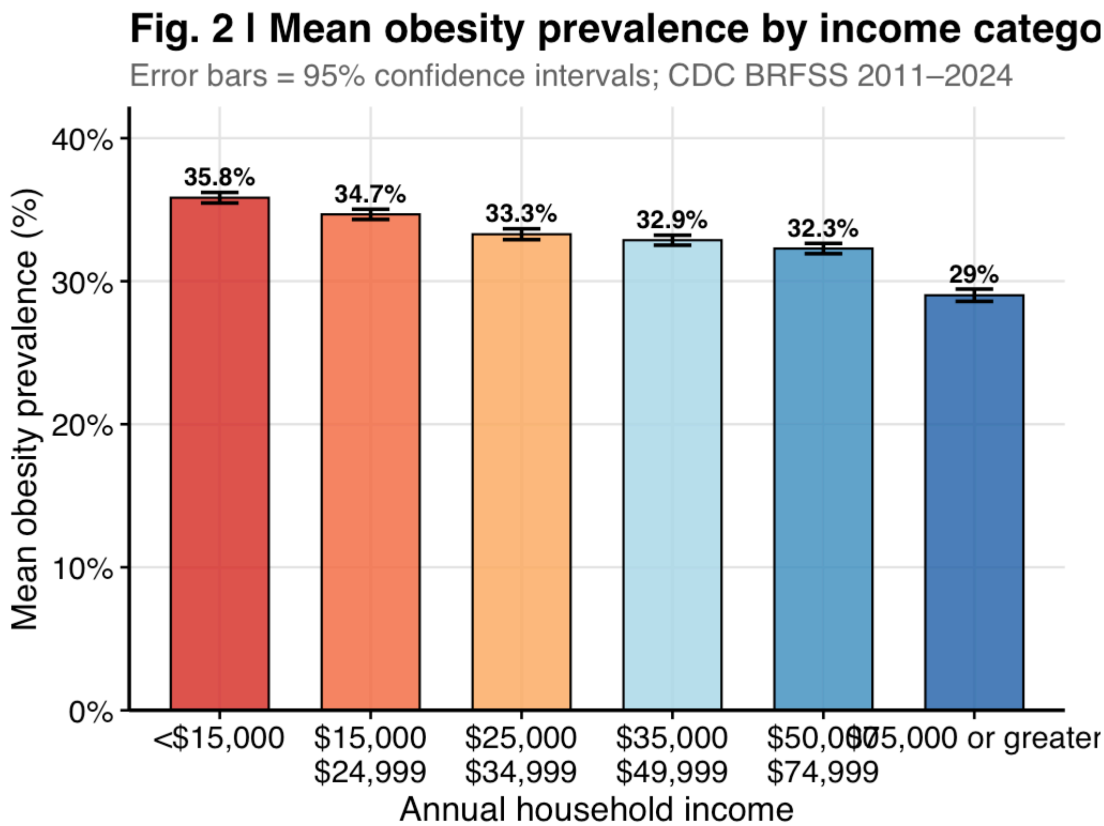

Figure 2: Fig. 2 | Mean obesity prevalence by income category. Error bars = 95% confidence intervals.

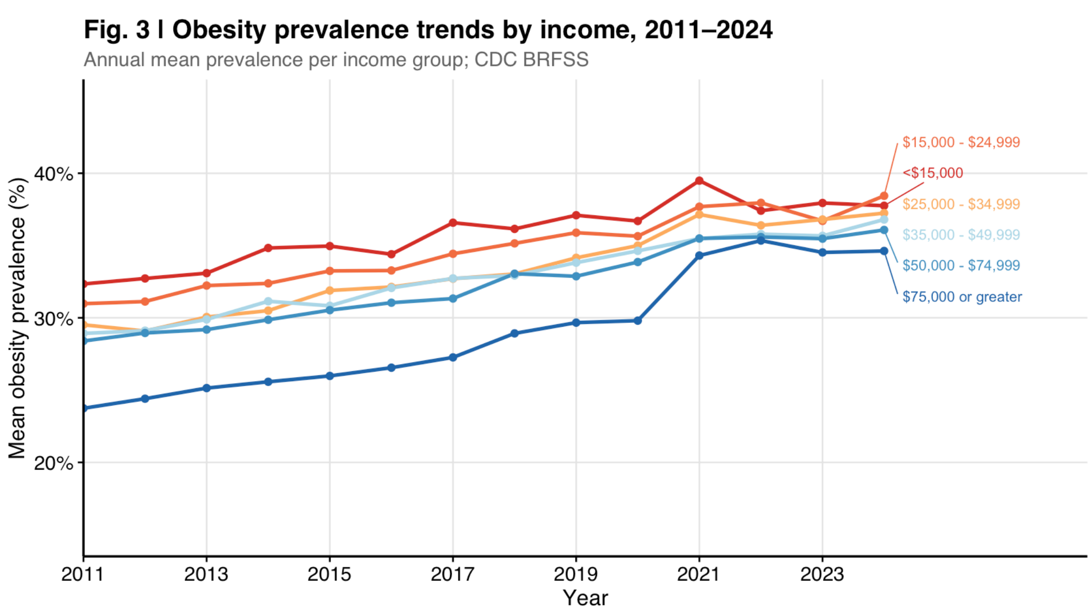

Figure 3: Fig. 3 | Annual mean obesity prevalence by income group, 2011–2024.

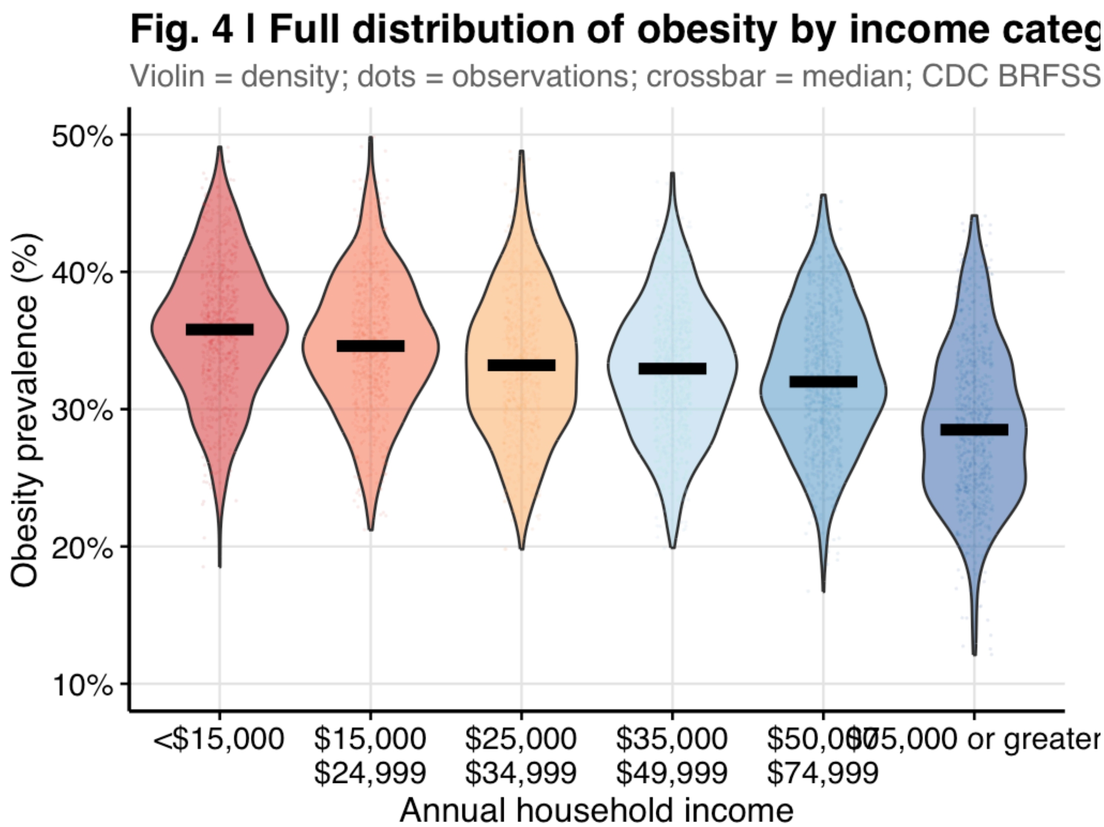

Figure 4: Fig. 4 | Full distribution of obesity prevalence by income category. Violin = density; dots = individual observations; crossbar = median.

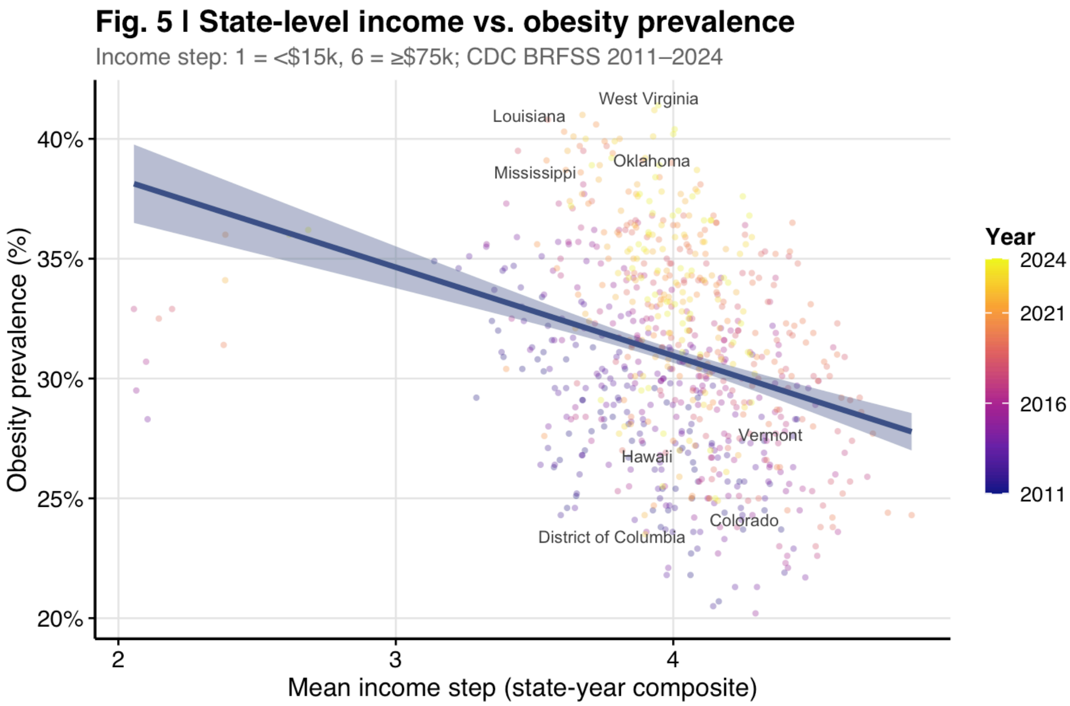

Figure 5: Fig. 5 | State-year obesity prevalence versus the sample-size-weighted mean income step (1 = lowest, 6 = highest).

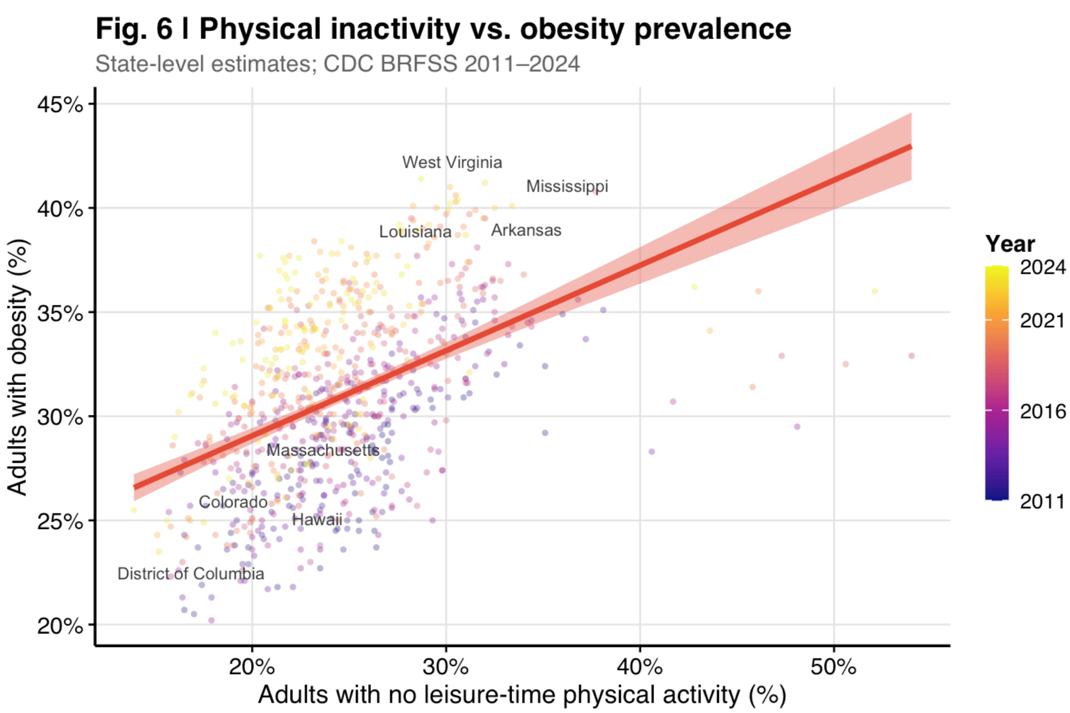

Figure 6: Fig. 6 | Physical inactivity vs. obesity prevalence across U.S. states (2011–2024). Each point is one state-year observation, coloured by year.

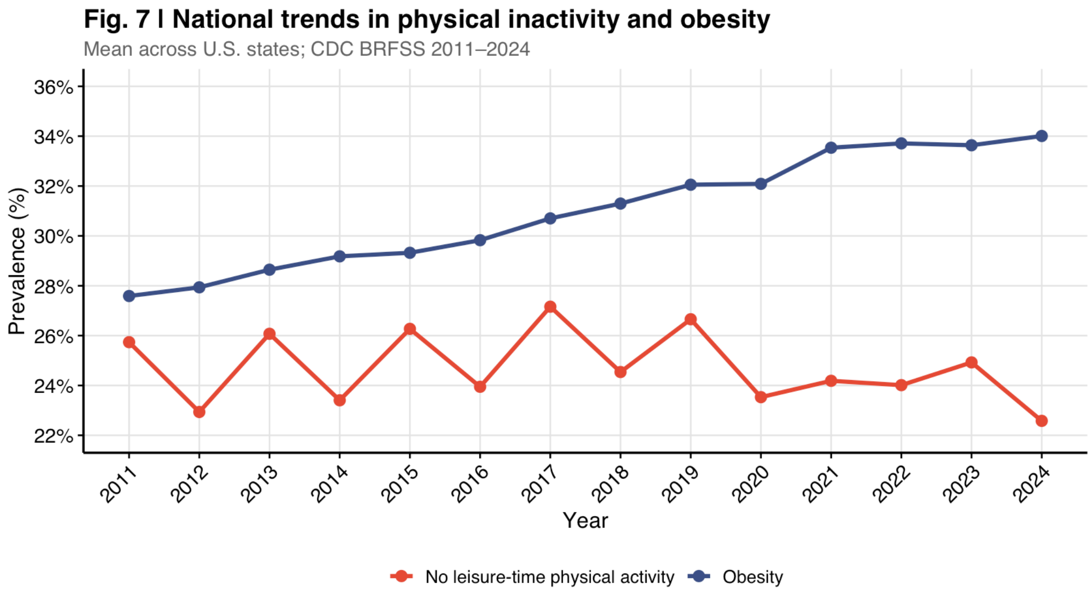

Figure 7: Fig. 7 | National trends in mean physical inactivity and obesity prevalence across U.S. states, 2011–2024.

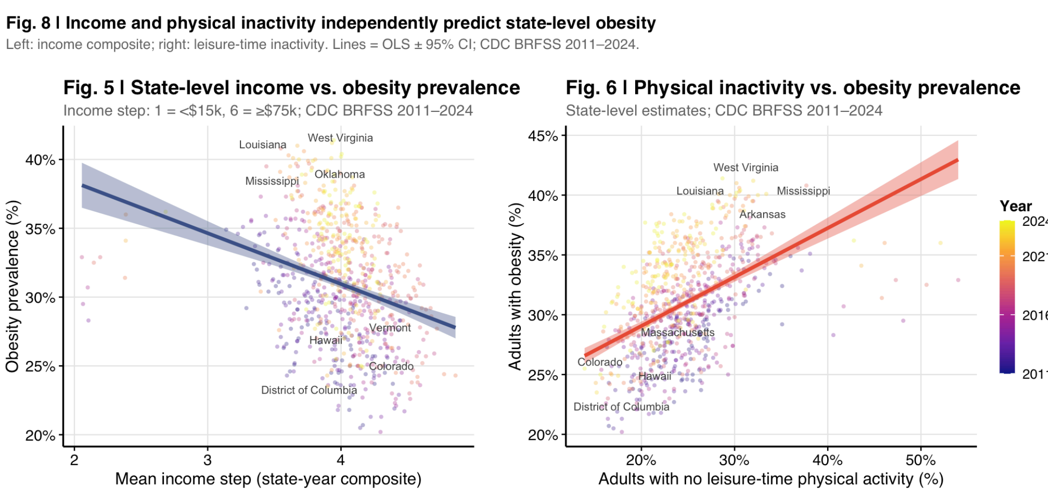

Figure 8: Fig. 8 | Income and physical inactivity independently predict state-level obesity. Left: income composite; right: leisure-time inactivity. Lines = OLS fit ± 95% CI.

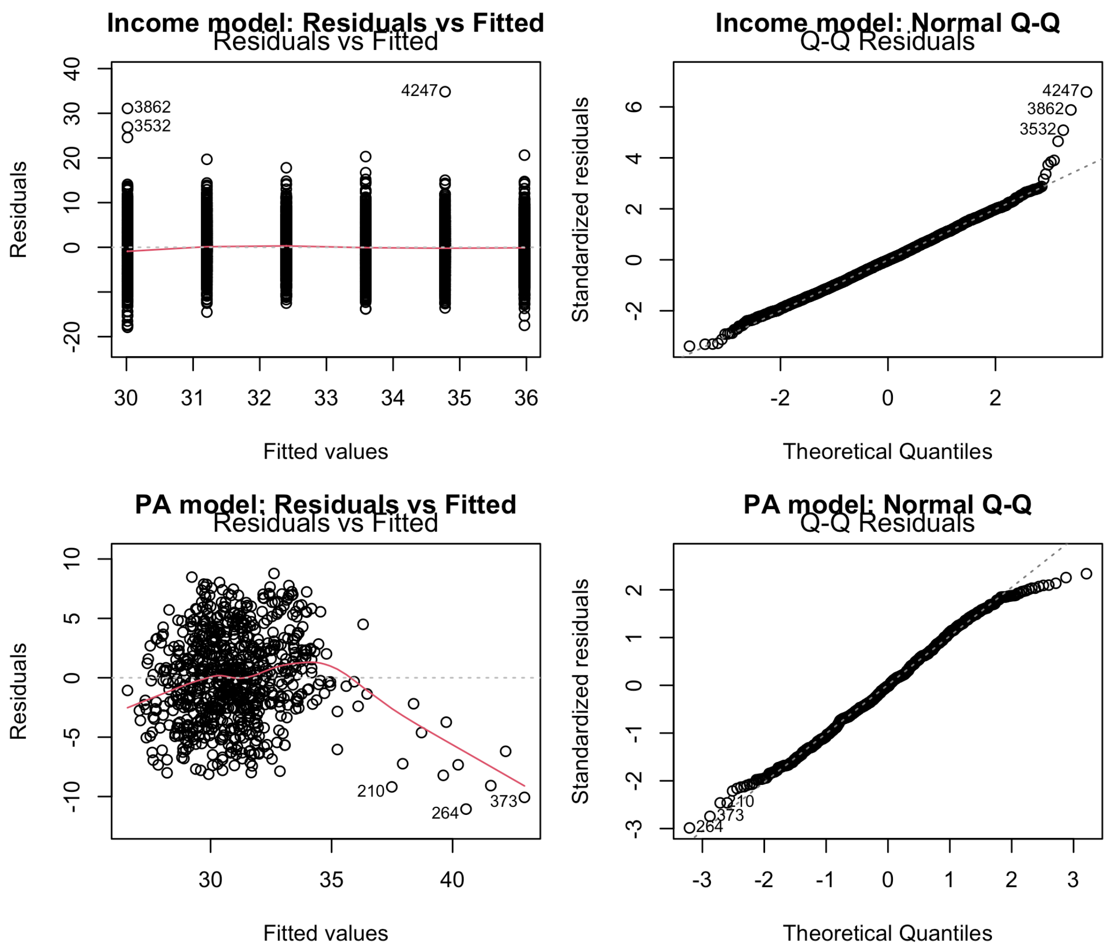

Figure 9: Regression diagnostic plots. Top row: income model. Bottom row: physical inactivity model.

Table 5: Simple linear regression: obesity prevalence (%) regressed on income step (1 = lowest, 6 = highest).

Term	Estimate	95% CI lower	95% CI upper	p-value
(Intercept)	37.162	36.809	37.514	0
income_step	-1.191	-1.281	-1.100	0

```
## R-squared     = 0.129
## Adj R-squared = 0.129
## RMSE          = 5.29
```

Each one-step increase in income category was associated with a **1.19** percentage point decrease in obesity prevalence (beta = -1.191, 95% CI: -1.281—1.1, p < 0.001). Income step explained **12.9%** of variance in obesity prevalence (R-squared = 0.129).

3.4.2 Q2 – State-Level Income Proxy and Obesity

```
## Pearson r = -0.304
## 95% CI:   [ -0.367 , -0.237 ]
## p-value:   <2e-16
```

Table 6: Simple linear regression: state-level obesity regressed on mean income step.

Term	Estimate	95% CI lower	95% CI upper	p-value
(Intercept)	45.719	42.388	49.049	0
mean_income_step	-3.692	-4.524	-2.861	0

```
## R-squared = 0.092
```

There was a significant negative correlation between the state-level income proxy and obesity (r = -0.304, 95% CI: -0.367—0.237, p < 0.001): states with a higher mean income composition had significantly lower obesity rates.

3.4.3 Q3 – Physical Inactivity and Obesity: Pearson Correlation

```
## Pearson r = 0.474
## 95% CI:   [ 0.416 , 0.528 ]
## p-value:   <2e-16
```

There was a significant positive correlation between state-level physical inactivity and obesity (r = 0.474, 95% CI: 0.416–0.528, p < 0.001), indicating a moderate-to-strong linear association.

3.4.4 Q3 – Simple Linear Regression

Table 7: Simple linear regression: state-level obesity prevalence (%) regressed on physical inactivity (%).

Term	Estimate	95% CI lower	95% CI upper	p-value
(Intercept)	20.882	19.507	22.256	0

Term	Estimate	95% CI lower	95% CI upper	p-value
pct_inactive	0.409	0.354	0.464	0

```
## R-squared     = 0.225
## Adj R-squared = 0.224
## RMSE          = 3.76
```

Each one percentage point increase in physical inactivity was associated with a **0.409** percentage point increase in obesity prevalence (beta = 0.409, 95% CI: 0.354–0.464, $p < 0.001$). Physical inactivity alone explained **22.5%** of variance in state-level obesity prevalence (R-squared = 0.225).

3.4.5 Q3 – Multiple Regression Adjusting for Year and Income

Table 8: Multiple linear regression: state-level obesity regressed on physical inactivity, year, and income proxy.

Term	Estimate	95% CI lower	95% CI upper	p-value
(Intercept)	-1142.456	-1250.079	-1034.833	0.000
pct_inactive	0.428	0.368	0.487	0.000
YearStart	0.577	0.524	0.630	0.000
mean_income_step	-0.325	-1.163	0.512	0.446

```
## Adjusted R-squared = 0.515
```

After adjusting for both secular time trends and the state-level income proxy, the association between physical inactivity and obesity remained **highly significant** (beta = 0.428, 95% CI: 0.368–0.487, $p < 0.001$). The adjusted R-squared of 0.515 indicates that the three-predictor model explains substantially more variance than either single-predictor model, confirming that income and physical inactivity both contribute independently to state-level obesity.

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4 Conclusions

This analysis of 2011–2024 CDC BRFSS data provides consistent ecological evidence that **both income and physical inactivity independently predict obesity prevalence** across U.S. states.

Income gradient (Q1 & Q2). A clear, monotonically decreasing income gradient in obesity was observed: mean prevalence decreased from 35.8% in the lowest income group to 29% among those earning >= $75,000. Each one-step increase in income category was associated with approximately a 1.19 percentage point reduction in obesity ($p < 0.001$). At the state level, states with a higher proportion of high-income residents had significantly lower obesity rates ($r = -0.304$, $p < 0.001$). These findings align with Ogden et al. (2017) and Drewnowski & Specter (2004), who attribute the income–obesity gradient to differential access to healthy food and recreational resources.

Physical inactivity (Q3). Physical inactivity was a significant positive predictor of obesity ($r = 0.474$, beta = 0.409, $p < 0.001$), explaining 22.5% of variance in state-level obesity prevalence, consistent with prior ecological analyses (Ding et al., 2016; Hill et al., 2003). Critically, this association **persisted after controlling for both year and the state income proxy** (beta = 0.428, $p < 0.001$), indicating that physical inactivity contributes to obesity risk above and beyond what income alone explains.

Joint interpretation. The multiple regression results answer the motivating question directly: a state with low obesity is characterised by *both* a wealthier population composition *and* higher physical activity levels –

these are independent, additive contributors. This aligns with Katzmarzyk et al. (2000), who showed that physical inactivity independently predicts obesity risk even after controlling for socioeconomic factors.

Limitations. This is an ecological analysis – state-level associations cannot be used to draw causal inferences about individuals (ecological fallacy). Both variables are self-reported, introducing recall and social desirability biases. The state income proxy is a derived composite rather than a directly measured construct. Unmeasured confounders – dietary patterns, food environment, urban density – may partially explain observed associations. Repeated measures within states over time are not formally modelled; a mixed-effects panel model would be more appropriate for causal inference.

Despite these limitations, the results reinforce the public health rationale for **dual-strategy interventions** targeting both income-linked food access disparities and structural barriers to physical activity as complementary pathways to reduce population-level obesity.

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5 AI Use Disclosure Statement

This document was produced with the assistance of **Claude (Anthropic)**. Claude was used to generate initial R code for data wrangling, visualisation, and statistical modelling across all three analytical blocks, and to assist with formatting the R Markdown document. All code was reviewed, tested, and adjusted by the author to ensure correctness and alignment with the assignment requirements.

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6 References

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